

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2022

Course Code: BIT101R01

Course: **BIOLOGY FOR ENGINEERS**

Question Paper No. : UGF018

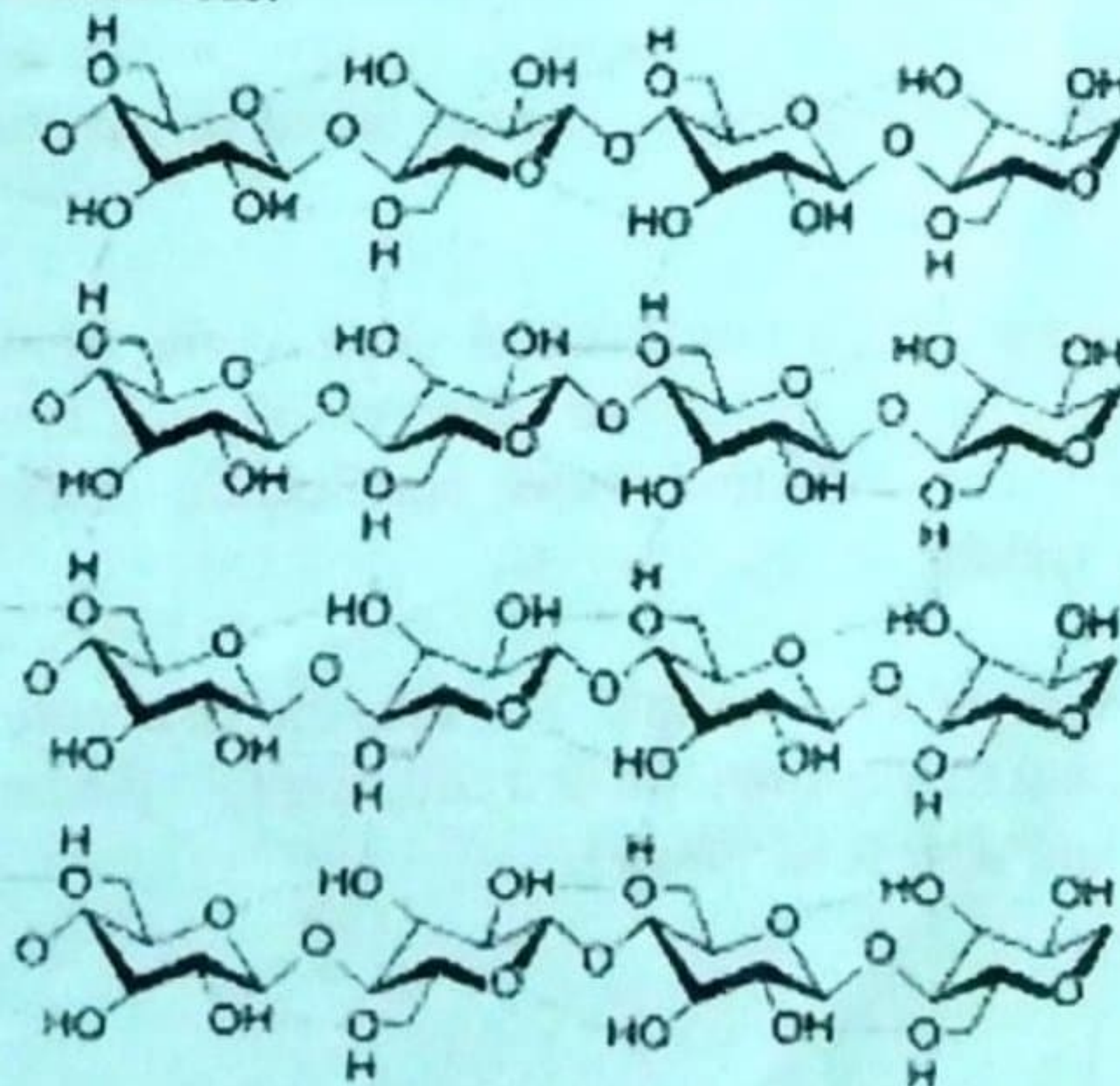
Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. A wound in mammals clot locally to prevent the blood from oozing out and regenerates the skin so as to heal the wound. Explain a product that has been inspired by this process to civil engineers in construction.
2. Carbohydrates are versatile compounds with several functional roles in Biology. Identify the given polysaccharide structure and mention its predominant role.



3. A smartphone is generally bought after looking at its features like cost, speed, display size, camera, build quality, etc. A company that sells smartphones need to optimize several of these features to compete in the market. Write a schematic of Genetic Algorithm for optimization of smartphone variants for this company.
4. Oxygen is one the most important molecule for the survival of heterotrophic organisms because it acts as a terminal electron acceptor and in the process, it generates ATP. On an average, we humans breathe >8 liters of air that helps us generate energy for our day-to-day activities. Distinguish the process of breathing from that of respiration.
5. Development of enzymes is one among many reasons that play a pivotal role for the emergence and evolution of life on this planet. List any four distinct features of enzymes that helps the maintenance and proliferation of cells.
6. Cardiovascular diseases (CVD) are the leading cause of death in the world. If given a chance, as an engineer, what kind innovation would you want to come up to reduce the incidence of CVD?
7. The futuristic transport is driverless cars, where the navigation is based on signals received by sensors of the car. Draw a schematic of learning and decision-making by the car based on Artificial Neural Network (ANN).
8. Safe potable water is the fundamental need of the human society. However, large populations of our society still lack access to safe water. Design a cost-effective water purification method inspired from the renal system.
9. Phospholipid bilayer is a simple but elegant molecule that gives individuality, be it to an organelle or a cell. Explain the uniqueness of this molecule and why it is robust?

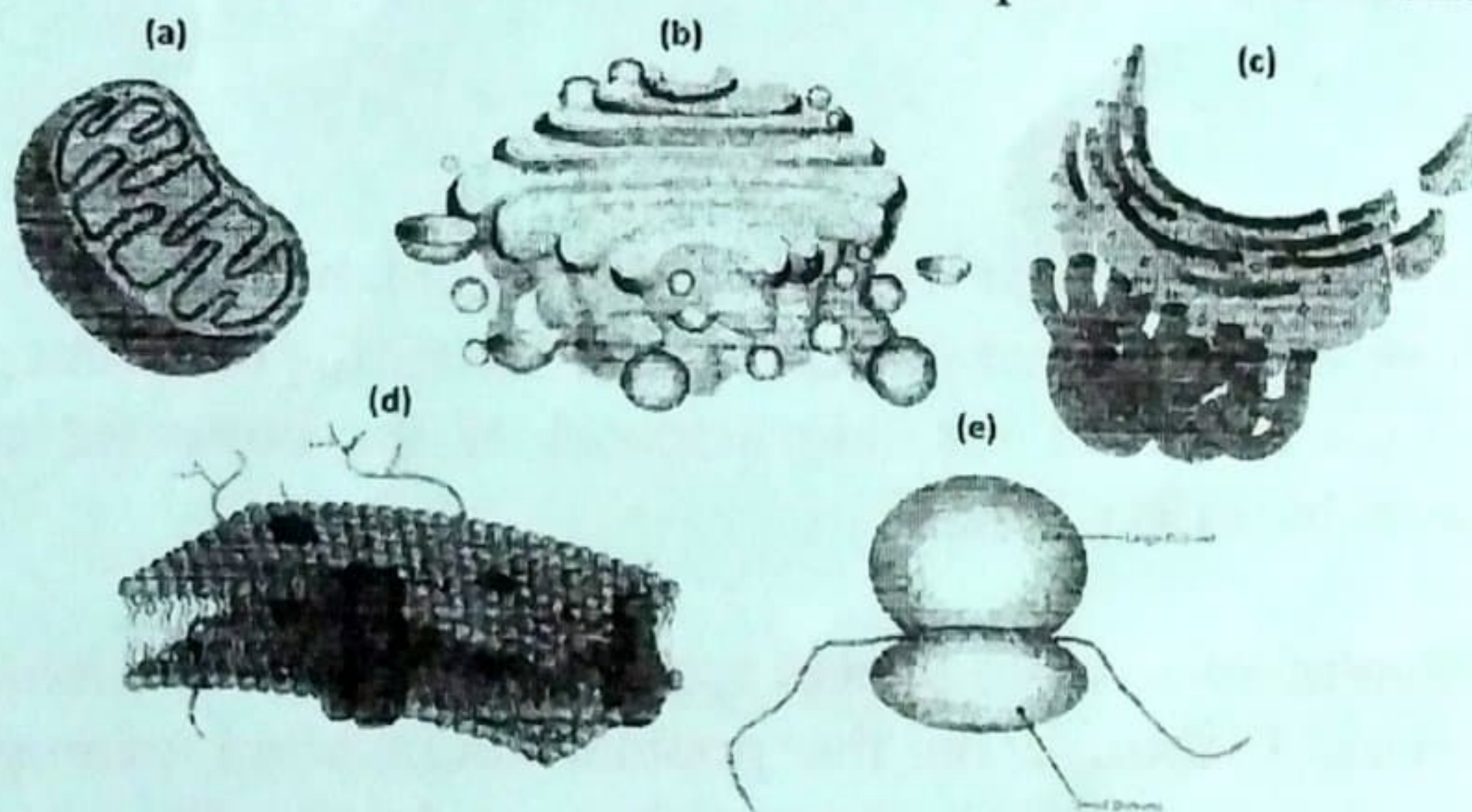
10. We are aware that COVID-19 pandemic is caused by Coronavirus and most of us got vaccinated against it. However, a variant of Coronavirus named Omicron has emerged and is causing fears of rampant spread, for which efforts to detect it are going on. Design a biosensor for detecting the Omicron.

PART – B

Answer all the questions

4 x 15 = 60 Marks

11. A cell is functionally viable due to the structural fidelity of biomolecules and the division of labor seen among the organelles. Identify the structure given below and explain its function.



(OR)

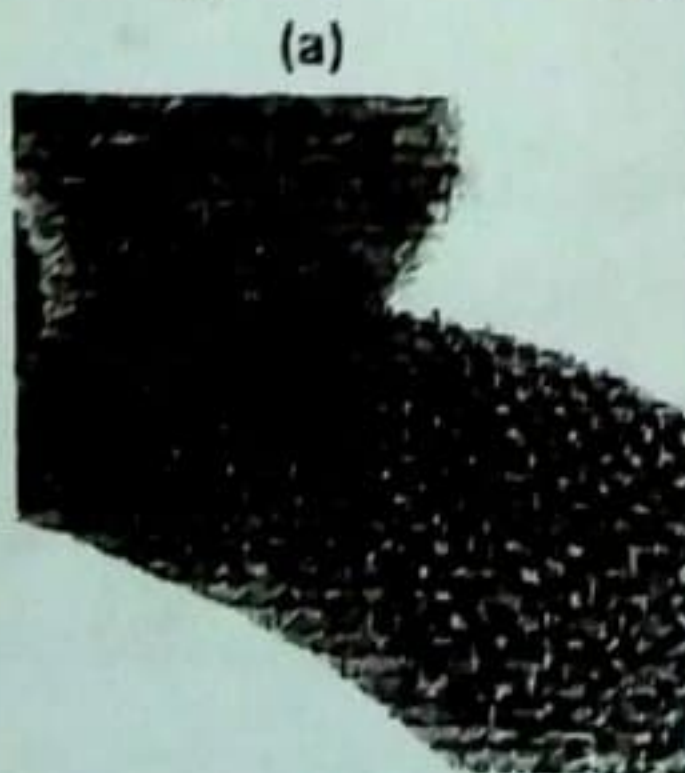
12. During the winters nowadays, Delhi and nearby areas face severe air pollution. One of the popular causes for this being the stubble burning in some northern states. From whatever we learnt in the course of 'Biology for Engineers', what do you think could be a potential solution to this problem? Draw a schematic and describe the process of cleaning the air.
13. List the components of the molecular toolbox that a cell needs to transcribe the information present in the DNA. Also, assume there are 40 amino acids in nature. What would then be the genetic code? Explain.

(OR)

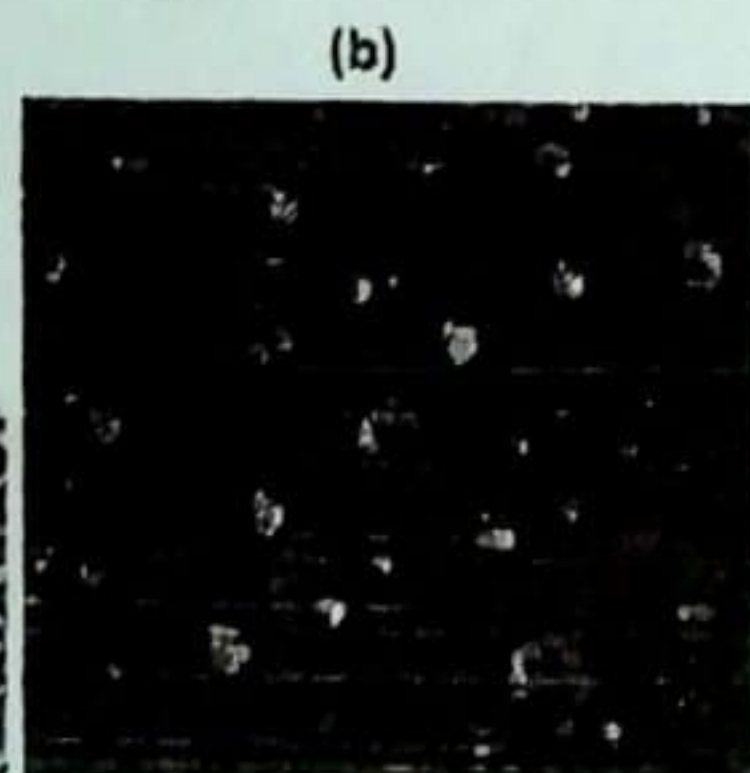
14. Design an early warning system to a car by obtaining bioinspiration from our innate immunity. Also, assume you are designing an anti-viral software, identify the features that can be learnt and implemented from our adaptive immune system?
15. The volume of data is growing exponentially necessitating newer strategies for storage, retrieving the stored data speedily, and keeping the data safely for a longer duration. DNA has a high potential of being a new storage medium. Explain the structure of the DNA molecule and describe why DNA has turned out to be a potential medium for storage.

(OR)

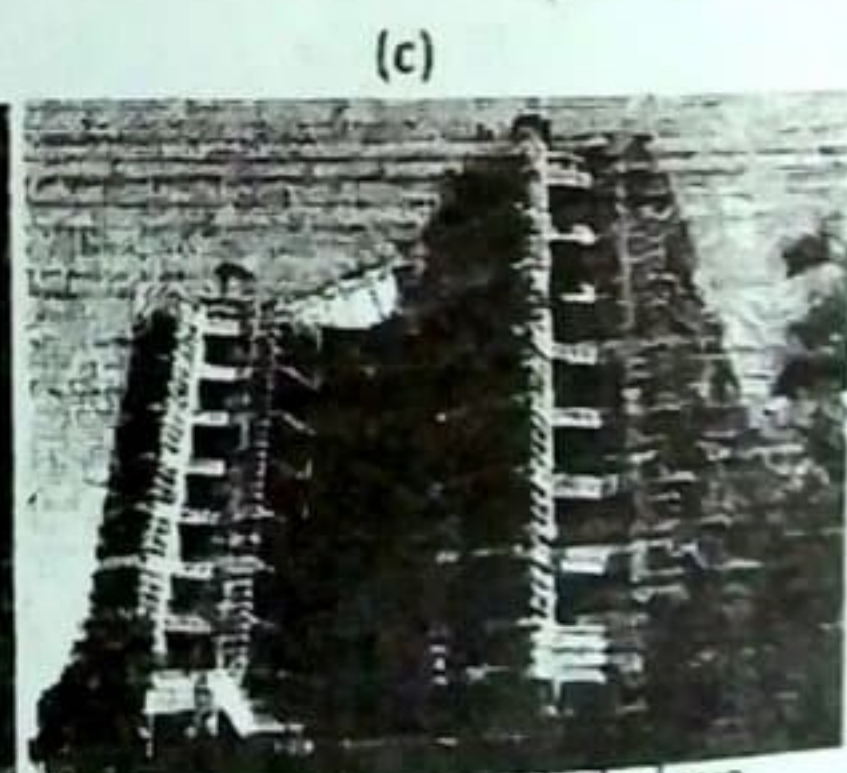
16. Assume that you must work in a biological laboratory for verifying a bioinspired design that you created. What are the different precautions that must be taken for safe disposal of the biowaste and explain different biosafety levels?
17. Creative/innovative engineers seek inspiration from Biology to solve problems. Elaborate on the popular bioinspired examples shown below, by mentioning the problem and how it was solved by observing the biological phenomenon.



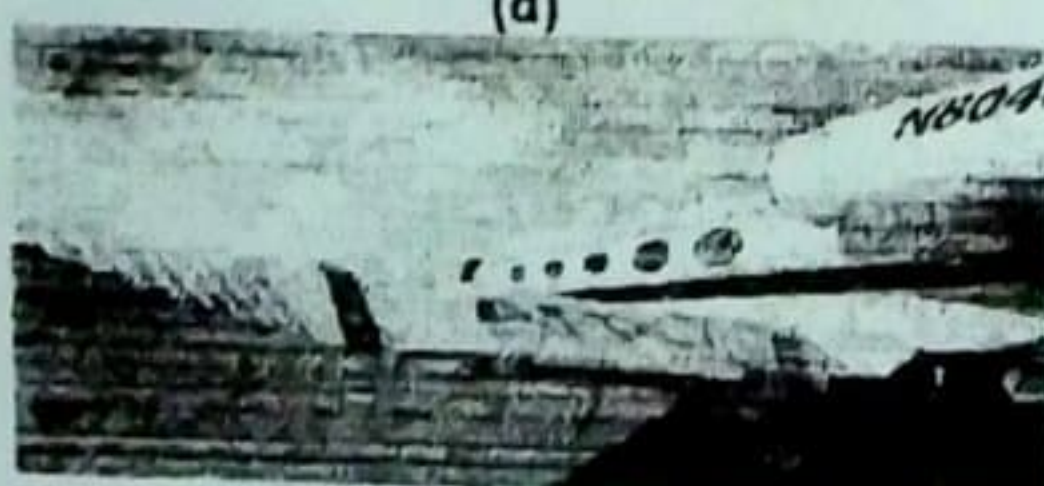
Velcro



Hydrophobic materials



East gate Centre, Zimbabwe



Twistable winged aircraft

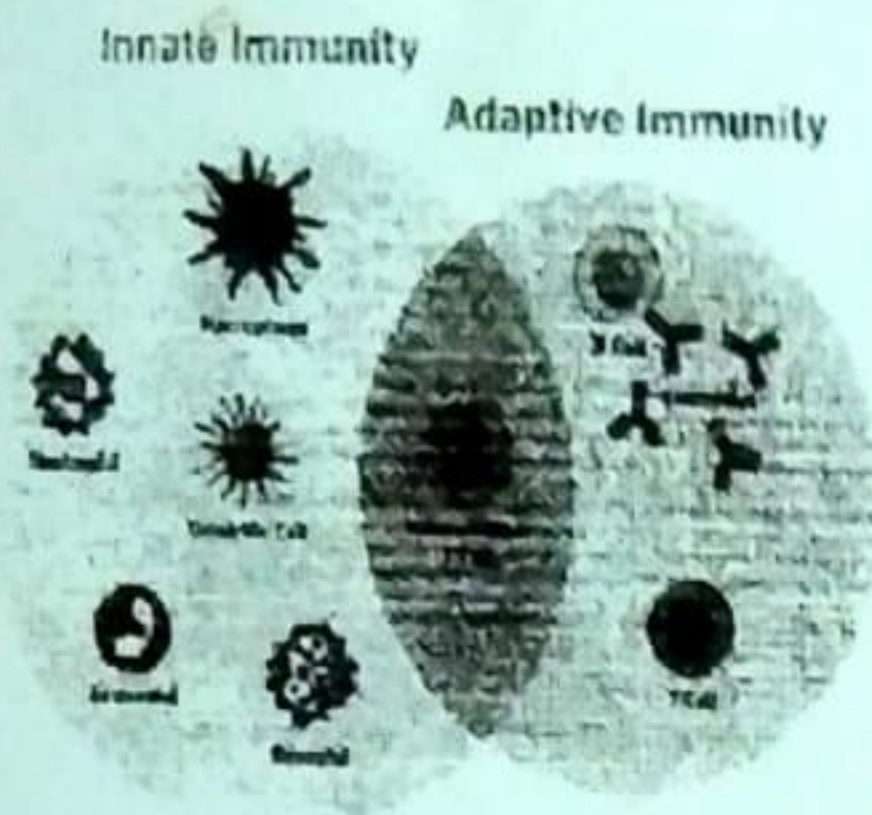


High-speed bullet train

(OR)

18. Creative/innovative engineers seek inspiration from Biology to solve problems. Elaborate on the biological system shown in the below image and how it can be taken as an inspiration for further engineering innovation.

(a)



Immune System

(b)



Swarm of birds

(c)



Nervous System

PART – C

Answer the following

1 x 20 = 20 Marks

19. In this pandemic, robot that could be supportive to the COVID positive patient is very much sought in a healthcare environment. As a student of Biology for Engineer's course, (a) envisage and define the task of the robot to be done in this context, and (b) appropriately design a bionic robot for this task.
